

Day 3: BIOS and POST Troubleshooting Theory

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Hardware or Configuration Issue
No Boot Device Found	The HDD/SSD is not detected, disconnected, or the boot order is incorrect.
CMOS Checksum Error	The CMOS battery may be weak/dead, or BIOS settings may be corrupted.
Keyboard Error	The keyboard may be disconnected, faulty, or not detected by the system.
Memory Error	RAM may be loose, faulty, or incompatible with the motherboard.
CPU Fan Error	The CPU fan may be disconnected, faulty, or spinning too slowly.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

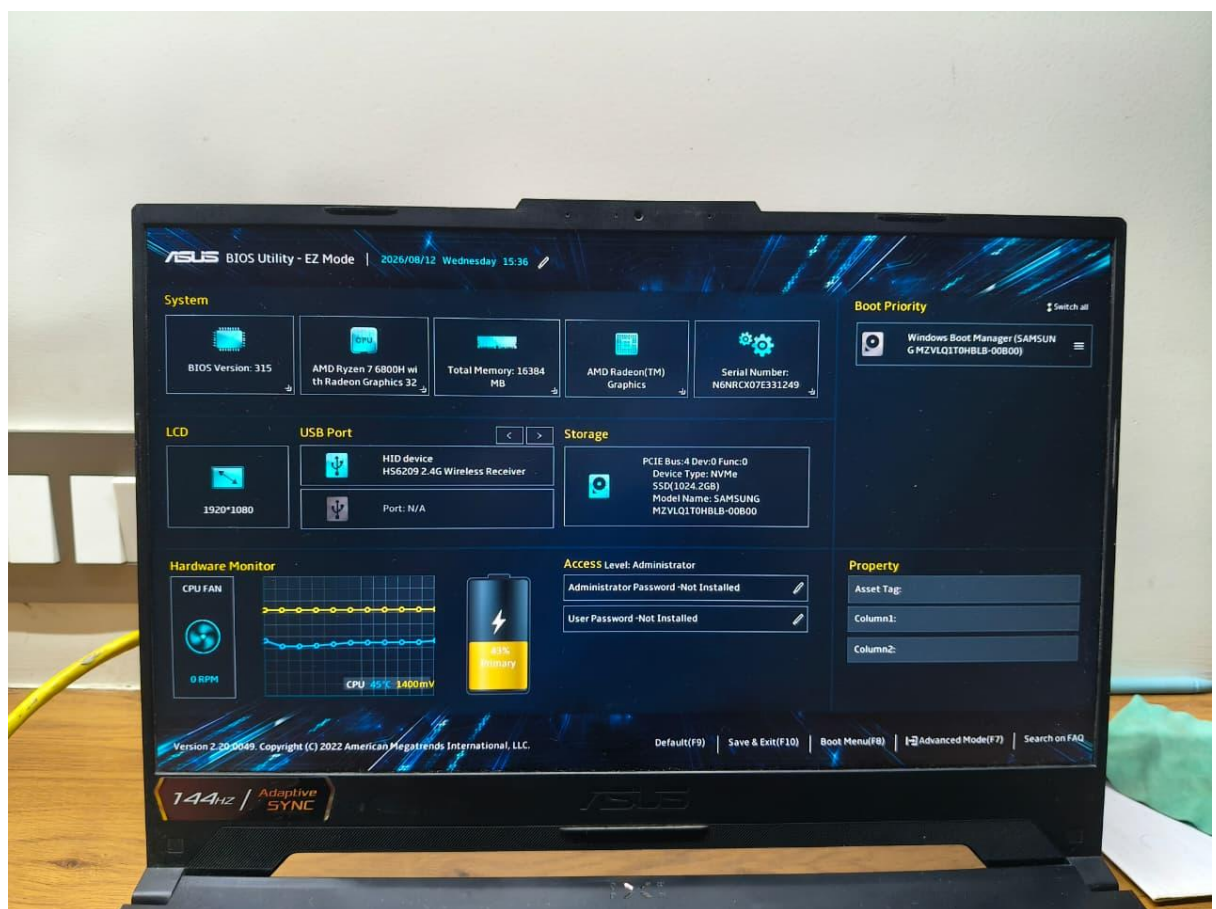
POST Beep Code Troubleshooting

Meaning: For a traditional **AMI BIOS**, **3 short repeating beeps** typically indicate a **base memory (RAM) read/write failure**. This means the system is having trouble detecting or testing the RAM during POST.

Two troubleshooting steps:

1. **Reseat the RAM:** Turn off and unplug the PC, remove the RAM modules, and firmly reinstall them in their slots. Make sure the locking clips are secure.

2. **Test the RAM individually:** If there are multiple RAM sticks, test them **one at a time** to identify a faulty module. You can also try a known-working compatible RAM module.
3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



Motherboard Manufacturer: ASUS
BIOS Version: 315
BIOS Interface: ASUS BIOS Utility – EZ Mode

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

CMOS Checksum Error – Defaults Loaded

Meaning:

This error means that the BIOS detected that the stored **CMOS/BIOS settings are corrupted, changed, or no longer valid**. As a result, the BIOS has loaded its **default settings** so the computer can start.

How to resolve it:

1. Enter the **BIOS setup** and check that the date, time, boot order, and other settings are correct.
2. Save the correct BIOS settings and restart the computer.
3. If the error appears again, **replace the CMOS battery** (usually a CR2032 battery).
4. Enter the BIOS again, set the correct settings, and save them.
5. If the problem continues after replacing the battery, check for a **BIOS/firmware problem or motherboard fault**.

Most common cause: A weak or dead CMOS battery.